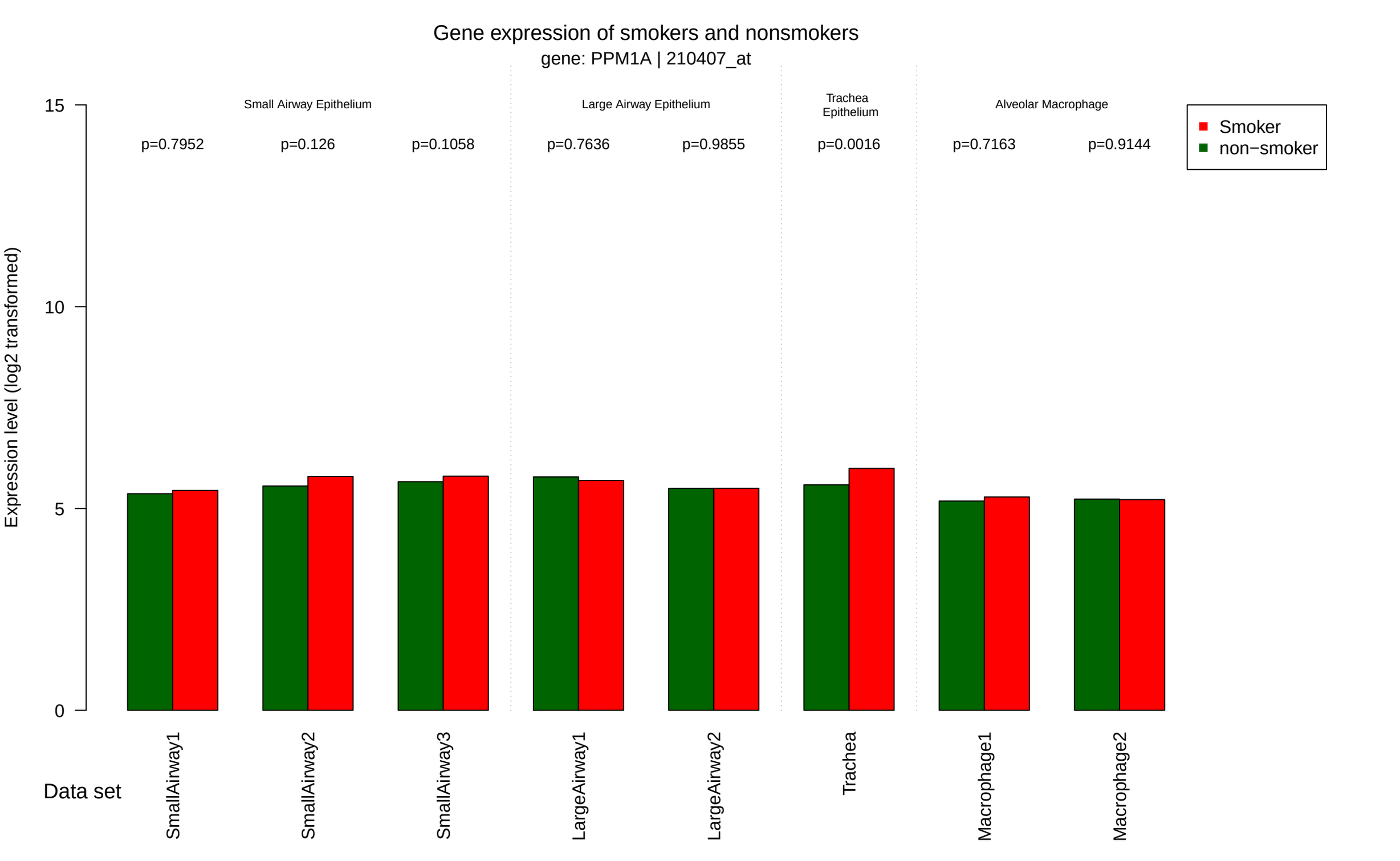
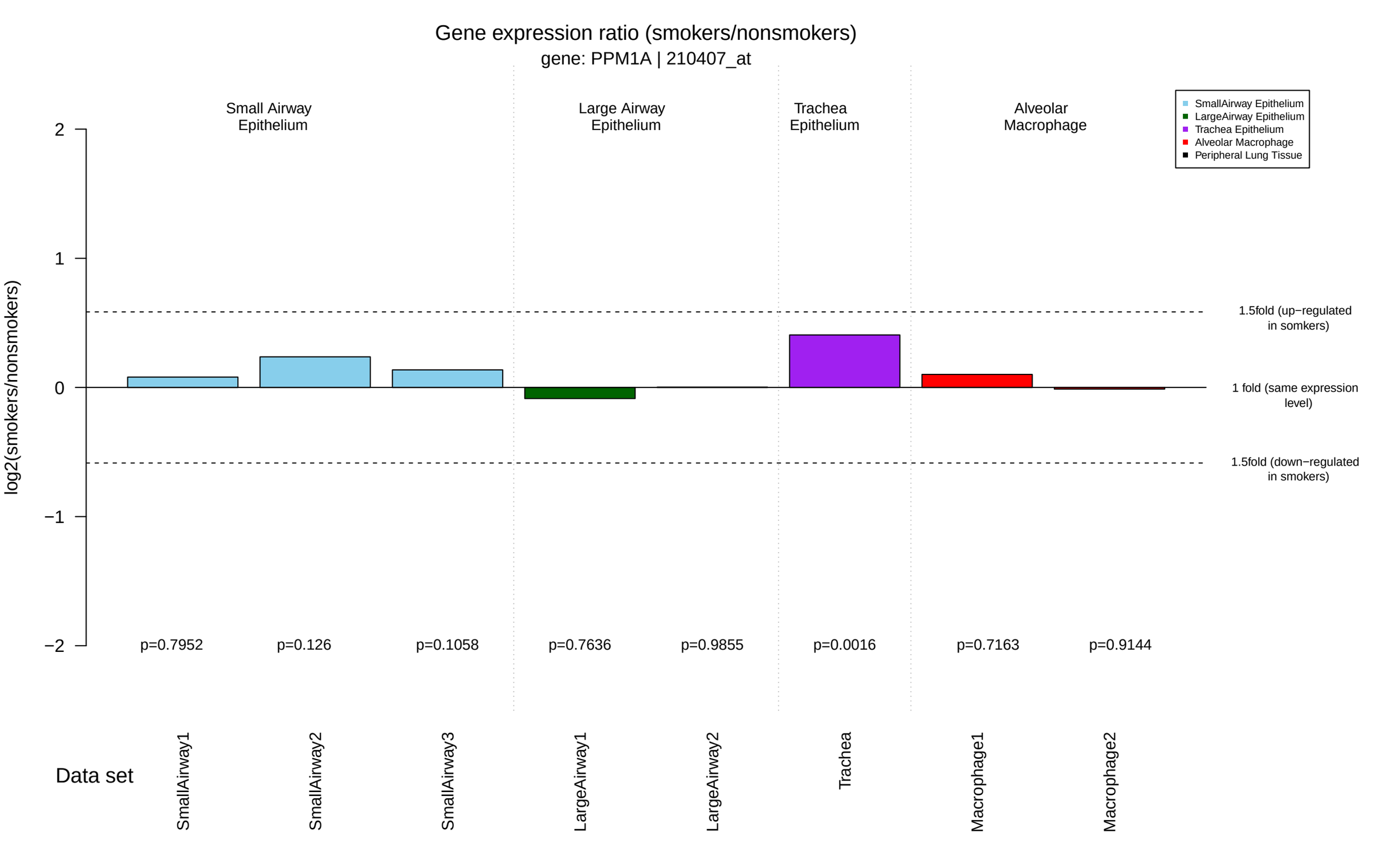
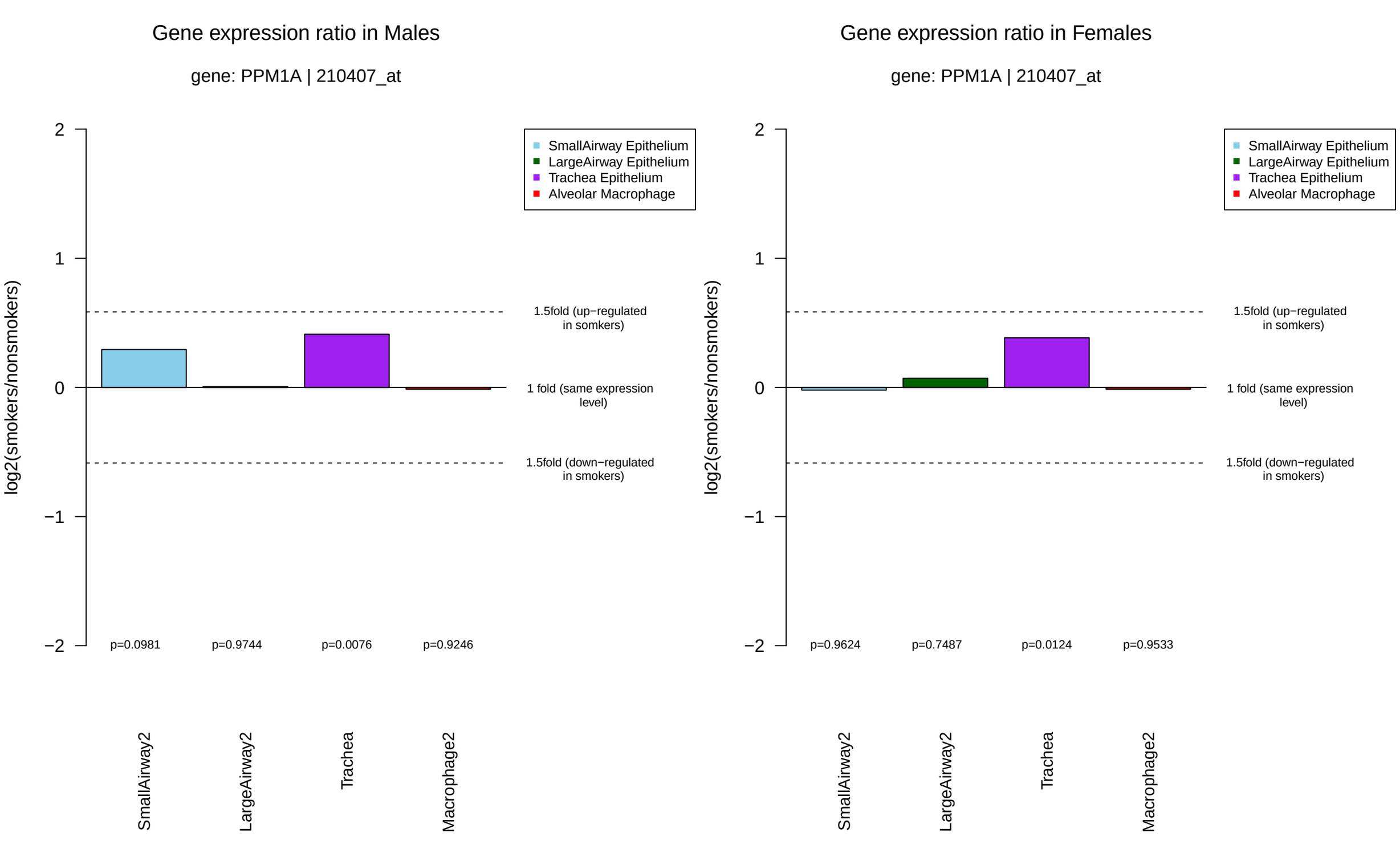
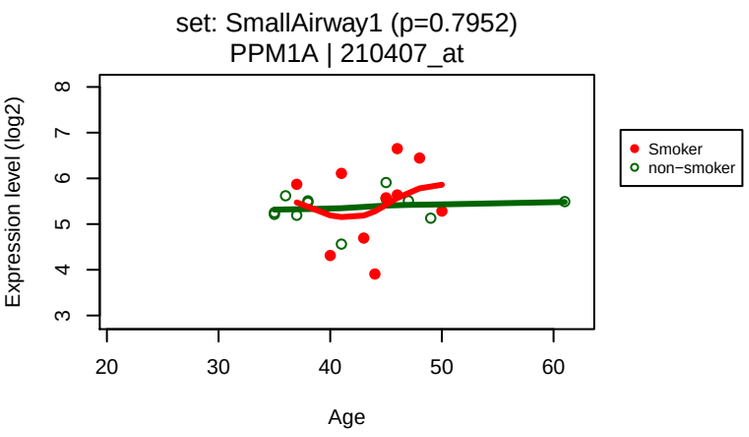
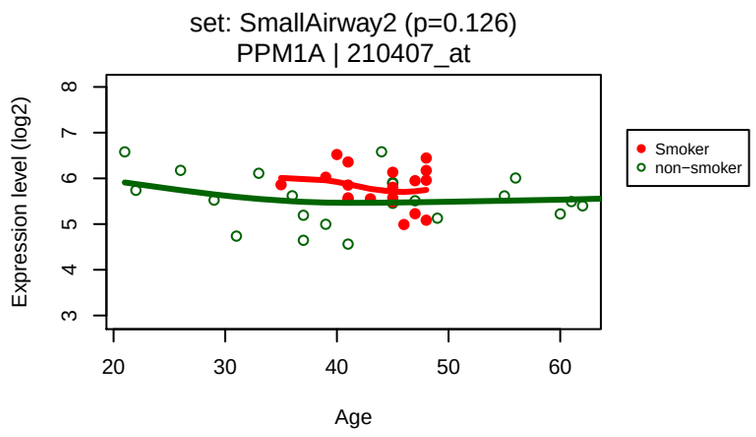
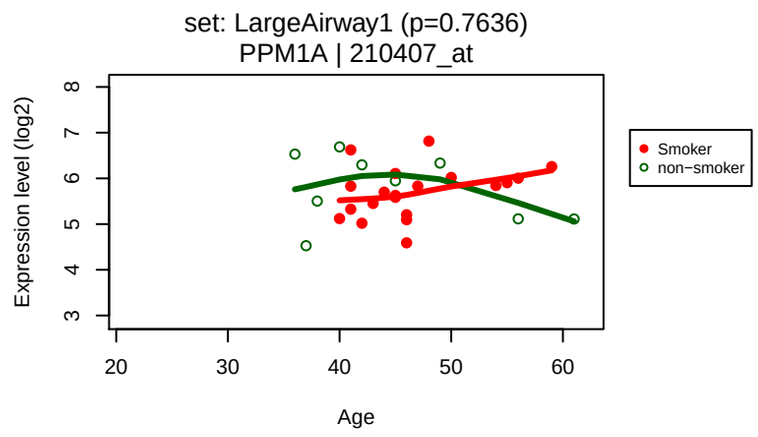

Gene expression of smokers and nonsmokers
gene: PPM1A | 210407_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7952
p=0.126
p=0.1058
p=0.7636
p=0.9855
p=0.0016
p=0.7163
p=0.9144
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

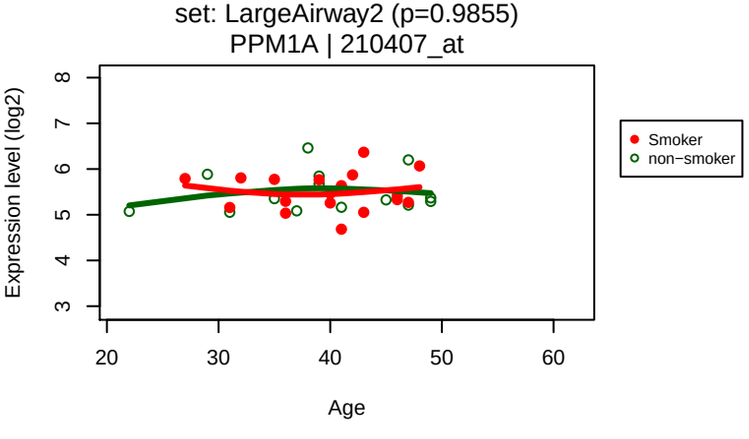

Gene expression ratio (smokers/nonsmokers)
gene: PPM1A | 210407_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7952
p=0.126
p=0.1058
p=0.7636
p=0.9855
p=0.0016
p=0.7163
p=0.9144
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

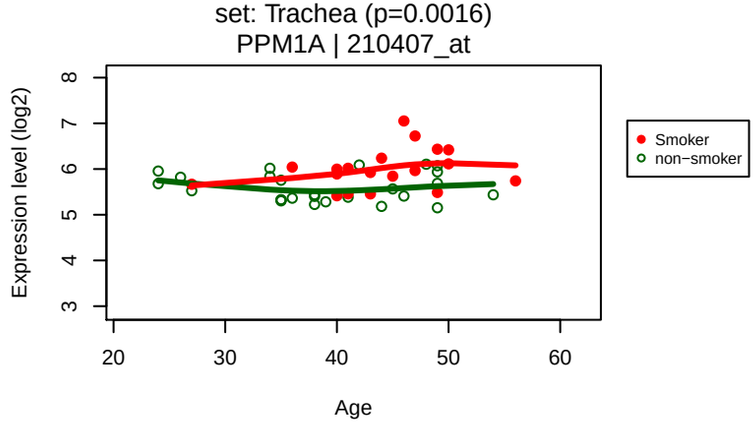

Gene expression ratio in Males
gene: PPM1A | 210407_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0981
p=0.9744
p=0.0076
p=0.9246
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PPM1A | 210407_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9624
p=0.7487
p=0.0124
p=0.9533
SmallAirway2
LargeAirway2
Trachea
Macrophage2

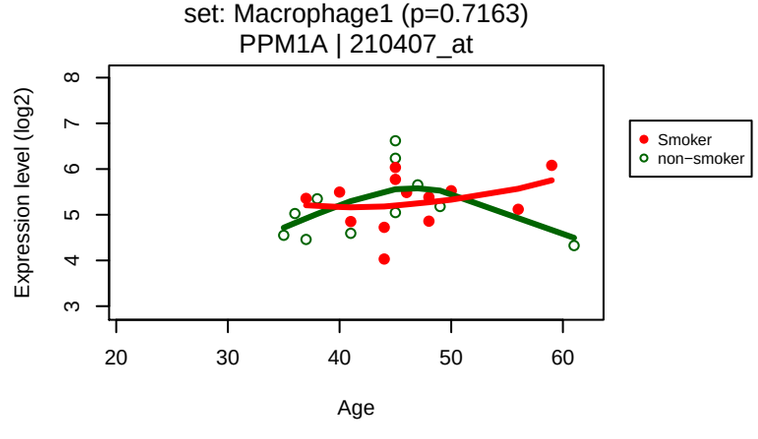

set: SmallAirway1 (p=0.7952)
PPM1A | 210407_at
Expression level (log2)
Age
Smoker
non−smoker

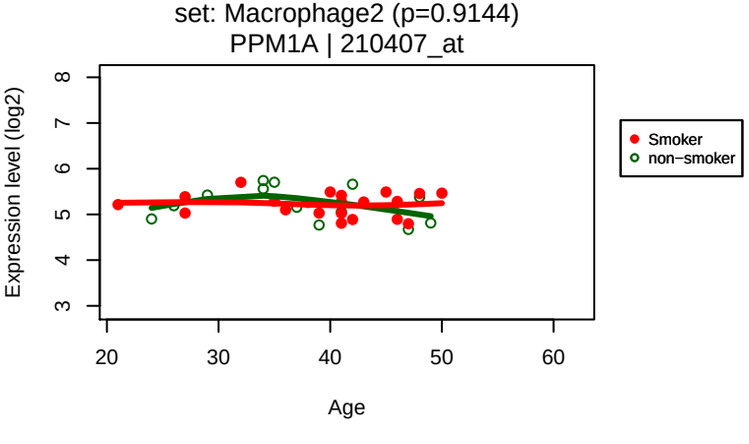

set: SmallAirway2 (p=0.126)
PPM1A | 210407_at
Expression level (log2)
Age
Smoker
non−smoker

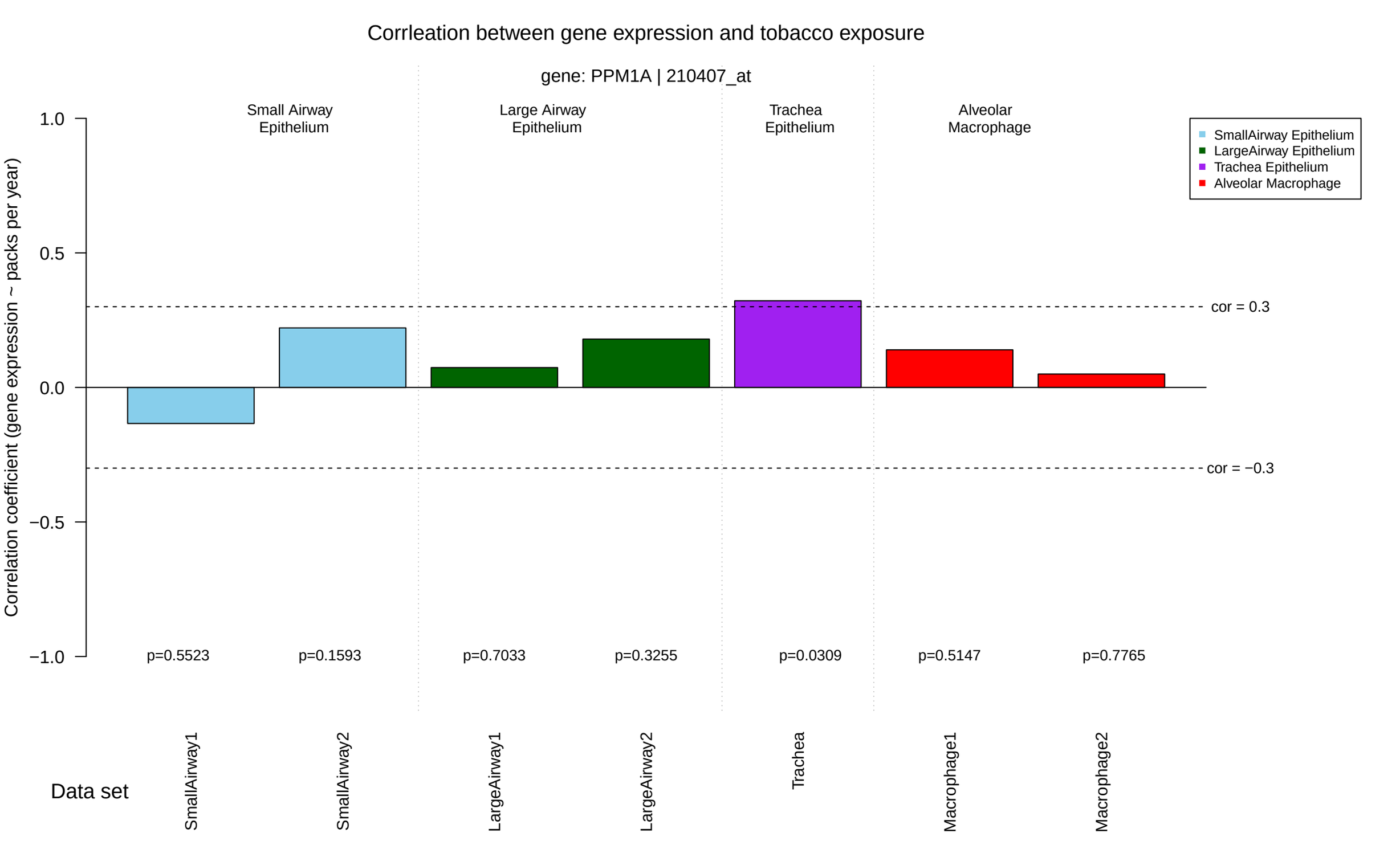

set: LargeAirway1 (p=0.7636)
PPM1A | 210407_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9855)
PPM1A | 210407_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0016)
PPM1A | 210407_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7163)
PPM1A | 210407_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9144)
PPM1A | 210407_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PPM1A | 210407_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.5523
p=0.1593
p=0.7033
p=0.3255
p=0.0309
p=0.5147
p=0.7765
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2